

Chapter 1 — Processes

A process is an instance of a program in execution. The OS represents each process via a Process Control Block (PCB) that holds the PID, register snapshot, memory map, scheduling state, and open-file table.

Processes transition through five canonical states: **new**, **ready**, **running**, **waiting**, and **terminated**. The dispatcher moves a ready process to running by performing a context switch.

Context switching is pure overhead — no user work is done during a switch. The cost depends on register count, cache footprint, and TLB invalidation.